

VARUN S

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Profile

Enthusiastic Computer Science Engineering student with a solid foundation in programming, data structures, and machine learning. Skilled in developing predictive models, deploying machine learning applications, and working with Python-based machine learning frameworks. Continuously strengthening expertise in machine learning and data science through structured learning and hands-on project experience.

Education

B.E - Computer Science Engineering, *Bannari Amman Institute of Technology* 2024 – Present
CGPA - 7.39/10

HSC, *KG Matriculation Higher Secondary School* 2023 – 2024
Percentage - 86.16%

SSLC, *KG Matriculation Higher Secondary School* 2021 – 2022
Percentage - 96.6%

Languages

- English
- Tamil

Skills

TECHNICAL SKILLS

- **Programming & ML Tools:** Python, SQL, C, Java, JavaScript, Streamlit, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Docker (basic)
- **Machine Learning & Data Science:** Supervised & Unsupervised Learning, Regression, Classification, Neural Networks, NLP (Masked Language Modeling), Data Preprocessing, Model Deployment
- **Computer Science Fundamentals:** Data Structures & Algorithms, Basic Backend Concepts
- **Version Control & Platforms:** Git, GitHub, Docker, Streamlit

Projects

Stroke Prediction Application – Python, Streamlit

- Designed an end-to-end machine learning pipeline for stroke risk prediction using structured clinical data.
- Performed data cleaning, feature engineering, and class imbalance handling to improve model robustness.
- Benchmarked baseline models against optimized tree-based models, achieving measurable gains in predictive performance.
- Deployed the trained model as a real-time inference web application using Streamlit, enabling interactive prediction on unseen data.
- Structured the project for reproducible deployment and future scalability.

Conditional Image Generation (cGAN) – Python

- Implemented a conditional Generative Adversarial Network to generate images conditioned on class labels.
- Analyzed generator–discriminator training dynamics, convergence instability, and mode collapse behavior.
- Evaluated output quality qualitatively and through loss-curve analysis to understand conditional generation effectiveness.
- Gained hands-on experience with adversarial training and deep learning optimization challenges.

Masked Language Modeling – Python

- Implemented masked token prediction to study contextual representation learning in language models.
- Built data preprocessing and masking pipelines to simulate pretraining objectives used in transformer-based models.
- Evaluated prediction accuracy on masked tokens to understand language modeling behavior and limitations.

Learning Management System – Python, Node.js, Docker

- Contributed to backend logic and frontend integration for a multi-user learning management platform.
- Implemented basic containerization using Docker to enable consistent development and deployment environments.
- Gained exposure to full-stack application flow and deployment considerations.

Certificates

- Python for Data science & Analytics(Hp)
- Machine Learning using python(Simplilearn)
- Microsoft Azure
- AWS cloud practitioner essentials

Problem solving

- Solved ~100 Data Structures & Algorithms problems on LeetCode
- Comfortable with Easy-level problems; progressing to Medium-level
- Focused on improving time & space complexity optimization

Activities

- Active member of Code Circle (competitive coding club)
- Participated in DSA and algorithmic problem-solving contests

Declaration

I declare that the information provided in this resume is accurate and true to the best of my knowledge. I am willing to provide further details or clarification if required.